Data Science Lab in Medicine - Big Data in Health Care

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Lab 2 - The Cox regression model

We will work on a dataset from a controlled clinical trial on N = 476 women with advanced ovarian cancer. The objective is to compare three different treatment regimens: - monochemotherapy P (cisplatin) - polichemotherapies, cyclophosphamide (CP), - cyclophosphamide+adriamycin (CAP) Endpoint is the overall survival. The prognostic factors available are residual tumor size, Karnofsky Index, Histology and the Federation Internationale de Gynecologie et d'Obstetrique (FIGO) classification.

Variables:

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npaz:      identification
tures :    residual tumor size (1 = absent, 2 = micro, 3 = ≤ 1cm,  4 = 1-2cm, 5
           = 2-5cm, 6 = 5-10cm, 7 = > 10cm)
kar:       Karnofsky Index (kar performance measure for rating the  ability of
           a person to perform usual activities, 20, 30,…,100)
histo:     Histology (isto=1, 2,…,6)
figo:      Federation Internationale de Gynecologie et d'Obstetrique
           classification with stages I, II, III
treat:     treatment (1=CAP,2=CP,3=P)
surv:      event indicator (1=death, 0= censoring)
survtime:  time to event (days)
```

Lets read the data and recode the variables tures, kar and isto:

tures

0	1	2
≤ 2 cm	2-10 cm	>10 cm

kar

0	1
≤ 70	> 70

histo (serous or not)

0	1
1, 7	2-6

Question 1.

Estimate the KM survival function for each treatment

Question 2.

Test the null hypothesis of equality of the survival curves, by using nonparametric tests

Question 3.

Test the null hypothesis of equality of the survival curves (specific for treatment), by using nonparametric hypothesis tests stratifying for the prognostic factors

Question 4.

Fit the Cox model including the treat covariate

Question 5.

Fit the Cox model including the treat covariate and adjusting for the other prognostic factors

Question 6.

From the Cox model in 5. estimate and plot: 1. the baseline survival curve 2. the survival curve for a patient treated with CP having the following characteristics: FIGO= III, Karnofsky Index =80, residual tumor size=3 cm, histotype=3

Question 7.

Compare the survival curve in 6. with the survival curves of two patients with the same characteristics, but treated with P and CAP, respectively

Question 8.

Fit the model of point 5. and check the PH assumption for karc and for histoc by the schoenfeld residuals plot

Question 9.

Check the PH assumption for karc and for histoc by the log(-log(Survival)) plot

Question 10.

Fit the model of point 5. but stratifying on the histoc variable

Question 11.

Fit the model of point 5. but with kar (numerical) instead of karc. Check the linearity assumption using the martingale residuals plot and splines

Question 12.

Fit the model of point 11. but with a time-dependent effect for histoc: include an interaction term with the function of time g(t)=ln(t)-ln(3)